

CHARACTERISTIC PARAMETERS OF MZ80 E18-D80NK - INFRARED OBSTACLE SENSOR

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MOTIVATIONS

The E18-D80NK adjustable infrared proximity sensor is an IR-based distance switch with an adjustable sensing range of 3–80 cm, suitable for short- to medium-range object detection. According to the manufacturer, the sensor is intended for applications such as robot interaction, collision detection, and proximity sensing, where fast and reliable non-contact detection is required.

Its simple digital output interface and adjustable sensitivity make it a practical choice for embedded systems. Since the sensing range and detection behavior depend on the selected sensitivity level and target surface properties, experimental characterization is required to understand its practical performance. For these reasons, the E18-D80NK was selected as the test sensor in this study.

METHODOLOGY

An E18-D80NK adjustable IR proximity sensor was characterized by varying its onboard potentiometer from 0–100%. The sensor operates based on reflective infrared sensing, in which an IR emitter transmits radiation and a phototransistor detects the reflected signal from the target surface. The received signal is processed by an internal comparator, producing a digital NPN normally-open output. Adjustment of the potentiometer modifies the comparator threshold, thereby controlling the effective sensing distance. Measurements were conducted using white (A4 paper) and black (non-perfect) planar targets to evaluate surface reflectivity effects.

The effective detection range was determined by gradually increasing the potentiometer within the minimum and maximum range of the white surface. After adjusting the potentiometer percentage with the equation $\frac{100(x - \min)}{(\max - \min)}$ where x is the measured maximum range, measurements were also taken for the black surface. For each potentiometer setting and target surface, measurements were taken 5 times to evaluate repeatability with the average of these 5 values, and the standard deviation was calculated from these measurements. Hysteresis was measured as the difference between the average detection distance during target approach and the average release distance during target retraction at the same potentiometer percentages. This was repeated for the black and white surfaces. Reliability was evaluated based on the consistency of the digital output during 5 repeated tests at fixed distances when the potentiometer was set for the maximum threshold for the white surface.

Sensor performance was evaluated in terms of effective range, standard deviation, hysteresis, and reliability, providing insight into the sensor's stability, response consistency, and practical detection behavior.

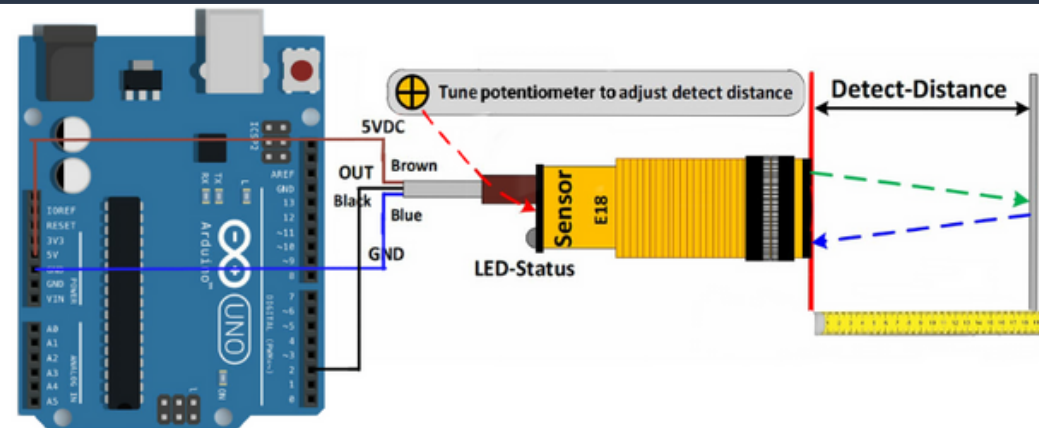


Figure 1: Testing Setup Diagram

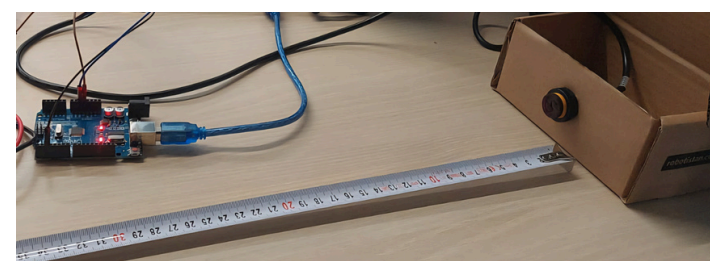
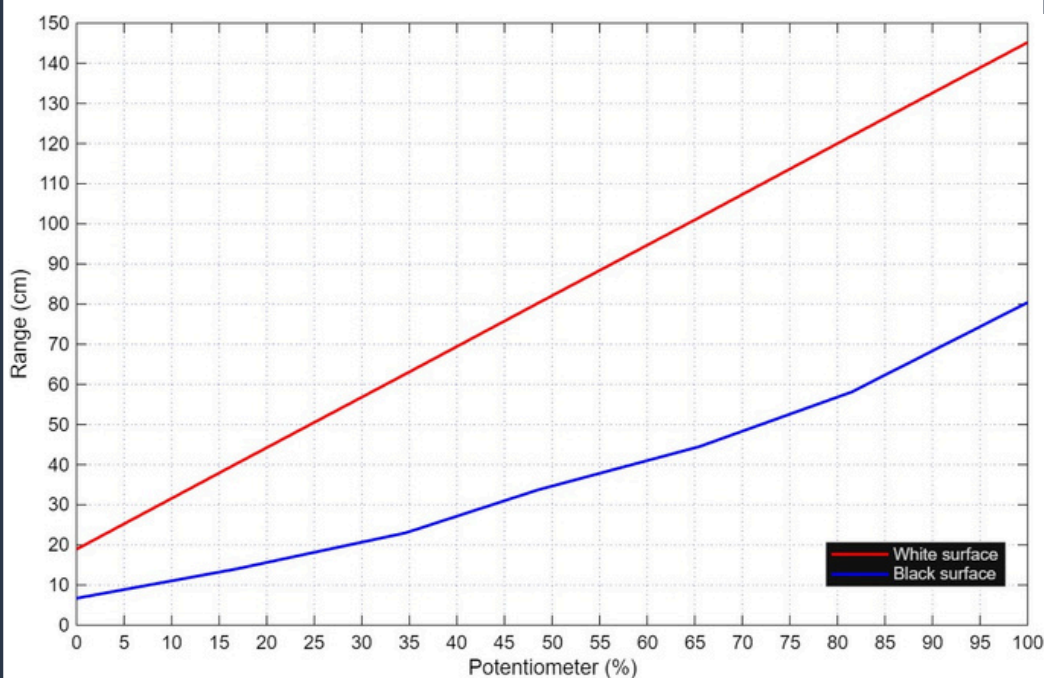


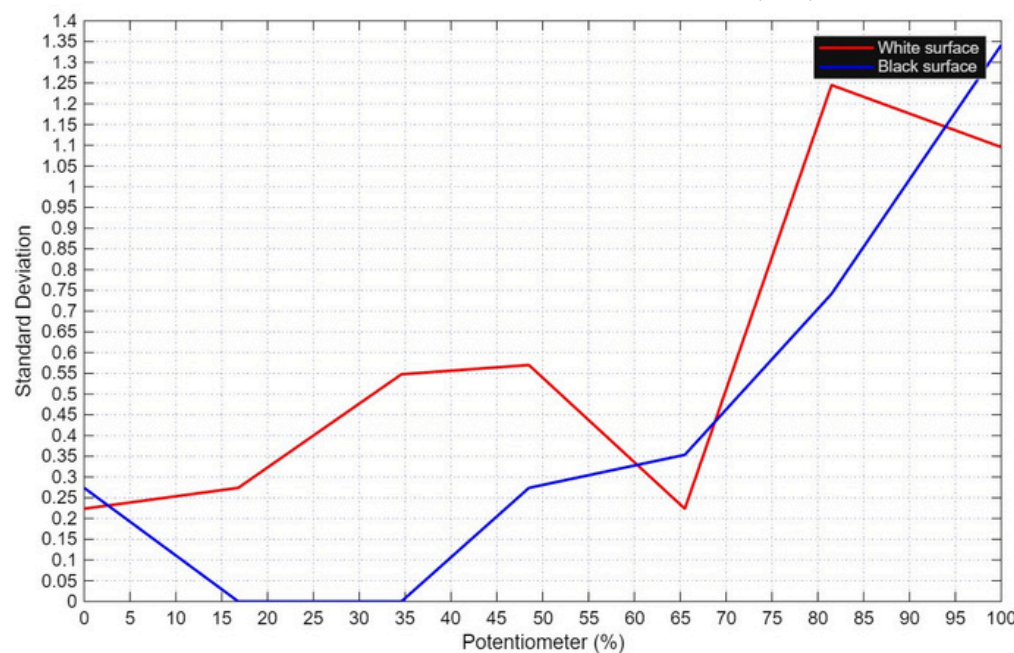
Figure 2: Testing Setup

Range - Potentiometer (%)



This graph shows the sensor's performance according to the potentiometer level. The sensor performs perfectly linear for the white surface as the potentiometer percentage was set according to it.

Standard Deviation - Potentiometer (%)



This graphic shows the repeatability of the sensor at the border of the maximum detectable range. The sensor loses its reliability as the potentiometer is scaled up. These numbers may not represent the complete behaviour of the sensor as the study only took 5 recorded measurements in a non-ideal environment. Lighting in the testing room, obstacle angle, human error can all count to the nonlinearity of this characterization.

SKETCH & FABRICATION

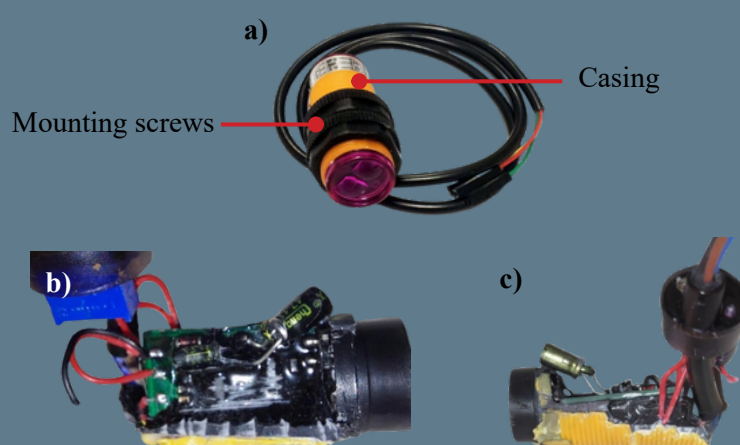


Figure 3: a) MZ80 E18-D80NK b) & c) MZ80 module exposed

MZ80 E18-D80NK is sold encased as seen in the above figures, which possesses the sensor module with potting epoxy poured over it. This provided a challenge to observe what is inside, and the vendor did not provide an image of the sensor module. However, other variations of MZ80 share a similar architecture with E18-D80NK.

Generic IR Obstacle Sensor Module

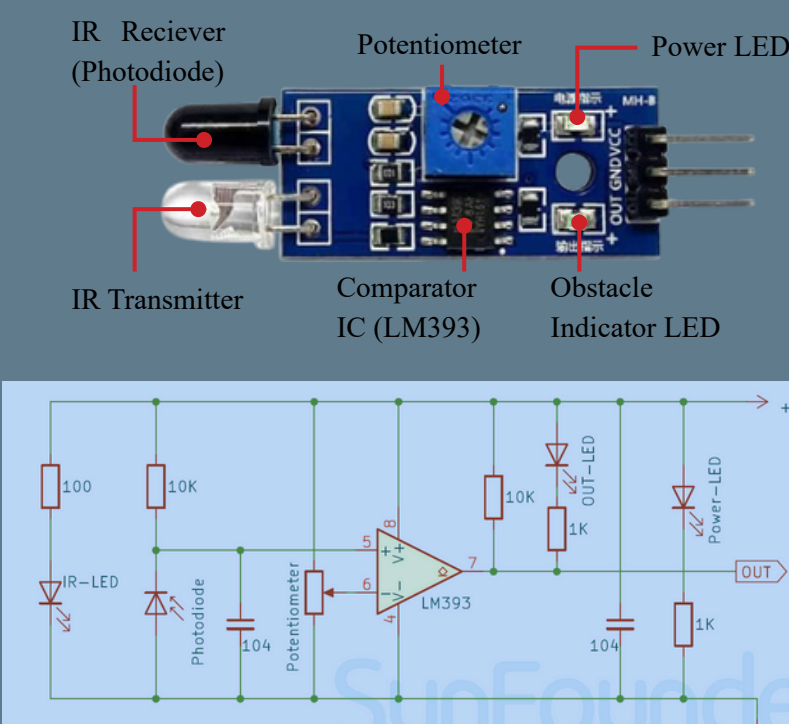
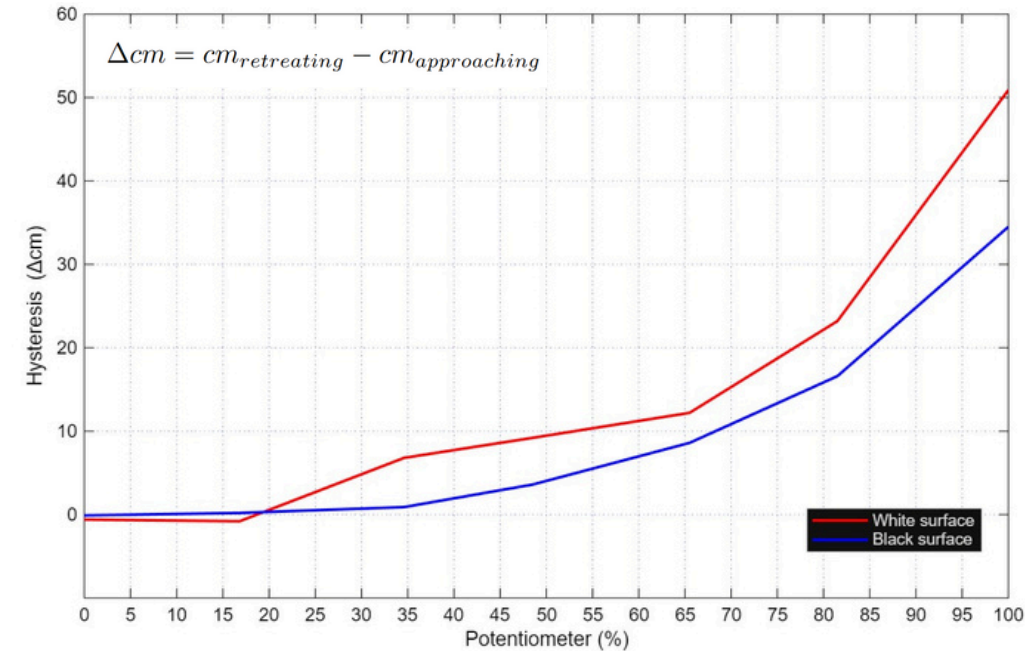


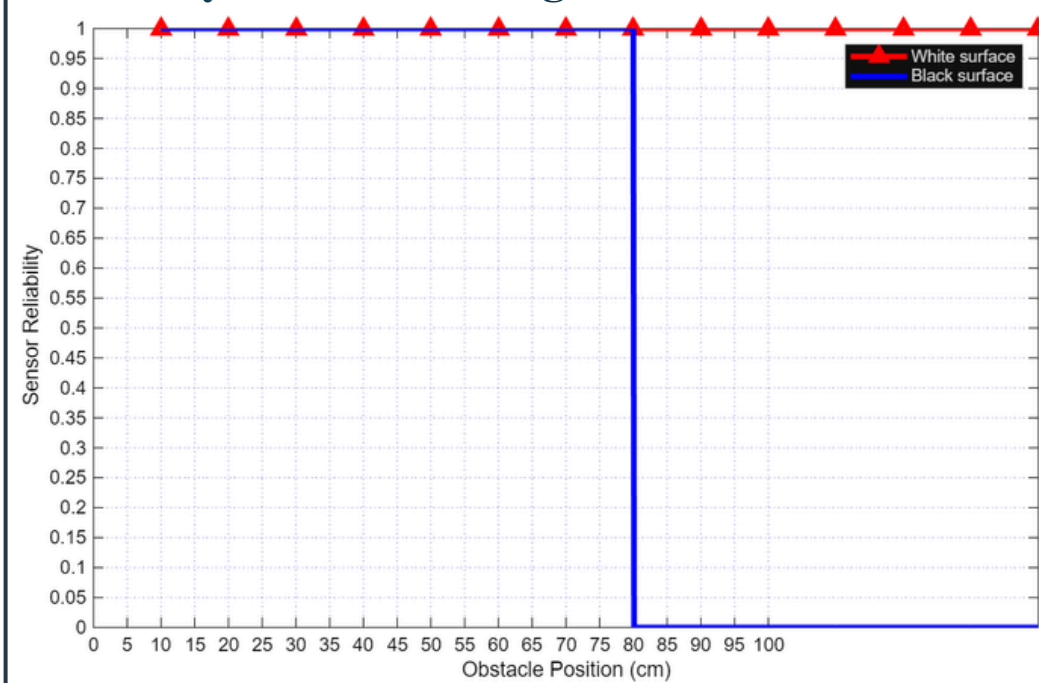
Figure 4: Generic IR Sensor Module Circuit Diagram(Sunfounder, n.d.)

Hysteresis (Δcm) - Potentiometer (%)



The sensor detection range presents differences according to the target's coaxial movement. This difference is documented with the equation within this graphic. The average of approach and retreat are subtracted and the results signify that the sensor has significantly increasing hysteresis as the potentiometer level is increased. The user must not use this sensor in a scenario if repeatable detection of an obstacle regardless of whether it approaches or not should not have any difference.

Reliability - Obstacle Range



This characterization follows a different method than the others. The potentiometer is scaled to the maximum so that the measurements are taken within the range and not at the border. Therefore, it can be said that this graph is always reliable to give the same result when it is within the range.

CONCLUSION

Experimental results indicate that the E18-D80NK infrared proximity sensor responds quickly and produces generally consistent detection distances across most potentiometer settings. At moderate sensitivity levels, the sensor provides reliable proximity detection, making it suitable for applications where fast object detection is required.

The experiments also highlight notable limitations. The potentiometer rotates continuously without a defined end stop, and beyond a certain adjustment point the sensor begins to detect distant background objects. In this region, the sensor effectively interprets the environment as a continuous reflective surface, leading to unintended detections and reduced range selectivity.

In summary, the E18-D80NK is well suited for simple and fast proximity sensing, but it requires careful manual calibration. The sensor is not intended for precise distance measurement and performs best when its effective sensing range is clearly constrained during use.

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